Human-Robot Affordance Dataset Annotation Pipeline

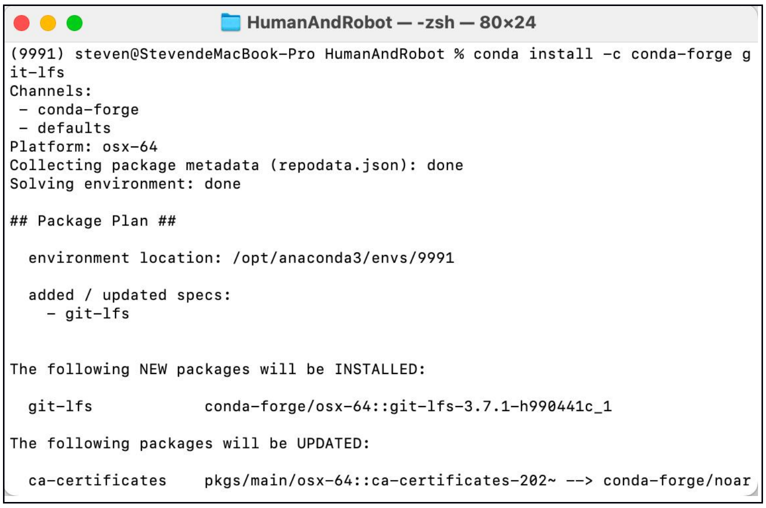
Section 1 Environment install

1. Install Anaconda:
https://www.anaconda.com/download

2. Create and activate a new conda environment:
conda create -n affordance python=3.10
conda activate affordance

3. Install required packages:
pip install h5py opencv-python numpy matplotlib

4. Install Git LFS:
conda install -c conda-forge git-lfs
git lfs install




```
(9991) steven@StevendeMacBook-Pro HumanAndRobot % conda install -c conda-forge g
it-lfs
Channels:
 - conda-forge
 - defaults
Platform: osx-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /opt/anaconda3/envs/9991

  added / updated specs:
    - git-lfs


The following NEW packages will be INSTALLED:

  git-lfs            conda-forge/osx-64::git-lfs-3.7.1-h990441c_1

The following packages will be UPDATED:

  ca-certificates    pkgs/main/osx-64::ca-certificates-202~ --> conda-forge/noar
```

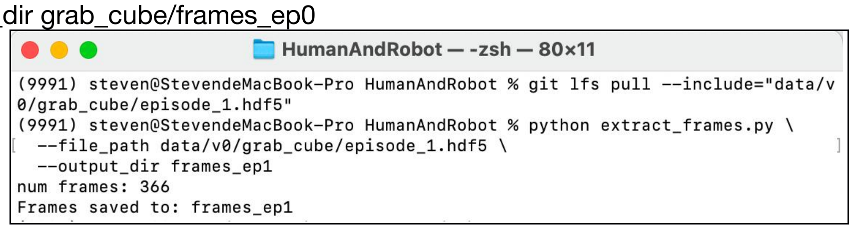
5. Clone the Human2Robot dataset repository without downloading the full dataset:
GIT_LFS_SKIP_SMUDGE=1 git clone
https://huggingface.co/datasets/dannyXSC/HumanAndRobot

6. Enter the repository:
cd HumanAndRobot, Loop 7 and 8.

7. Download only selected episodes (example):
git lfs pull --include="data/v0/grab_cube/episode_0.hdf5"

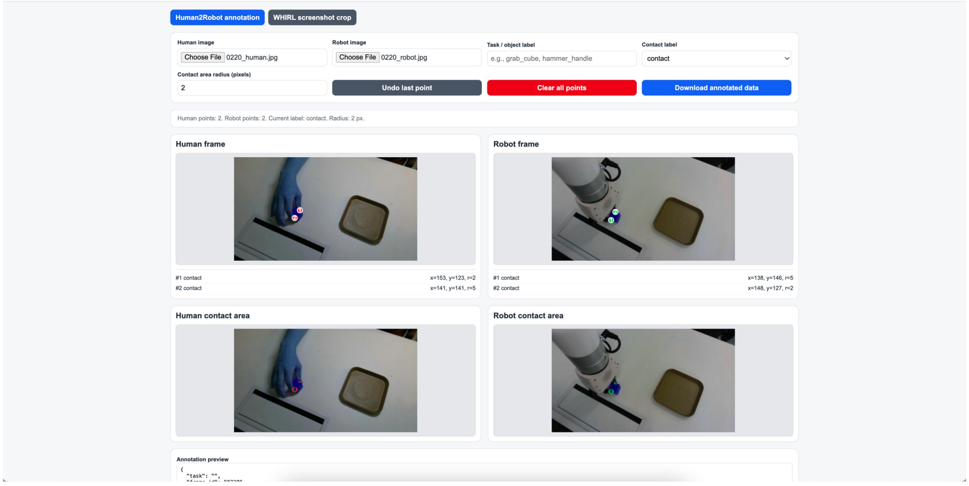
Do NOT run “git lfs pull” without “--include”, otherwise the full dataset (~233GB) may start downloading.

8. Extract paired human-robot frames:
python extract_frames.py \
  --file_path data/v0/grab_cube/episode_0.hdf5 \
  --output_dir grab_cube/frames_ep0




```
(9991) steven@StevendeMacBook-Pro HumanAndRobot % git lfs pull --include="data/v
0/grab_cube/episode_1.hdf5"
(9991) steven@StevendeMacBook-Pro HumanAndRobot % python extract_frames.py \
  --file_path data/v0/grab_cube/episode_1.hdf5 \
  --output_dir frames_ep1
num frames: 366
Frames saved to: frames_ep1
```

Section 2 Annotate contact points



9. Open “contact_annotation_tool.html” in a browser.

10. Load one paired human image and robot image into the annotation tool.

11. Annotate:
- human finger/object contact point
- corresponding robot gripper/object contact point
- test different area radius and choose the best match

The annotation should represent the same affordance semantics between human and robot interaction.

Do not annotate an occluded contact point by guessing. Do not force every grasp to have two points. One clear visible contact point is better than two noisy points.
For an edge contact point, annotate it. And set the radius=[1, 3].

Example:
human grasps hammer handle ↔ robot gripper grasps hammer handle

12. Export:
- Click “Download annotated data”

13. Annotation guidelines:
- use representative key frames only
- avoid blurred frames
- prefer clear contact interactions
- keep annotation consistent across tasks
- preserve original image filenames
- coordinates are stored in original image pixel coordinates

14.Key frames we need:
(1) start: The first frame of the task, human hand should be away from the target.
(2) pre_contact: A little frames before first_contact, human contact can look like touch the target;
(3) first_contact
(4) movement_1: Random frame after first_contact;
(5) movement_2: Another random frame after first_contact;
(6) placement:
(7) release